

1. OVERVIEW & SCOPE This policy outlines the protocols for using Generative AI tools and managing vector databases at Acme Innovation Labs. All engineering teams, data scientists, and contractors must adhere to these guidelines when deploying production RAG pipelines.

2. DATA CLASSIFICATION & EMBEDDING RULES - Public Data (Class 1): May be embedded using third-party hosted API models (e.g., OpenAI text-embedding-3-small). - Confidential Data (Class 2): Must be embedded locally using open-weights models (e.g., BGE-large-en-v1.5) on company-owned infrastructure. - Restricted Data (Class 3): Highly sensitive PII, source code, or financial records. Class 3 data MUST NOT be passed into vector databases under any circumstances.

3. VECTOR DATABASE HYGIENE - Index Refresh Rate: Production vector indices must undergo a full re-indexing cycle every 14 calendar days to remove stale or deleted documents. - Chunking Guidelines: Standard text documents must be chunked using a dynamic sliding window of 512 tokens with a 10% overlap (51 tokens). - Distance Metrics: All similarity searches across Class 1 and Class 2 collections must default to Cosine Similarity. Euclidean distance (\$L_2\$) is prohibited due to variance in high-dimensional embedding spaces.

4. RETRIEVAL & ACCURACY THRESHOLDS - Top-K Parameter: Vector queries must retrieve a minimum of Top-K = 5 chunks for general QA and Top-K = 10 for technical policy checks. - Similarity Cutoff: Any chunk returning a similarity score below 0.72 must be filtered out prior to passing context to the LLM prompt. - Fallback Procedure: If zero chunks meet the 0.72 similarity threshold, the RAG pipeline must return the standard system message: "I cannot find sufficient context in the knowledge base to answer this query."

5. COMPLIANCE & AUDITING All vector queries and retrieved chunk IDs must be logged in the centralized logging system (Axiom) for a minimum retention period of 90 days. Quarterly audits are conducted by the Security & Compliance Board on the first Monday of every quarter.